Chemical Engineering Journal 398 (2020) 125652

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

Supplementation of O_2-containing gas nanobubble water to enhance methane production from anaerobic digestion of cellulose

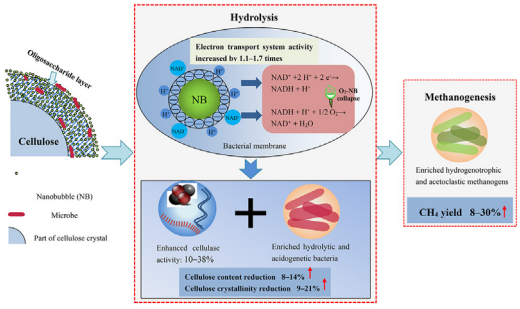
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HIGHLIGHTS

- NBW enhanced cellulase activity and cellulose crystallinity reduction during AD.
- O_2-NBW addition yielded 30% higher CH_4 production from cellulose than the control.
- O_2-NBW might provide a micro-oxygen environment achieving the highest ETS activity.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Anaerobic digestion
Crystal cellulose
Electron transport system
Methane
Microbial community
Nanobubble water

ABSTRACT

Nanobubble water (NBW) contains 10^6–10^8 gas bubbles per milliliter with diameter < 1 μm. These fine bubbles possess high mass transfer efficiency and hydrophobic property beneficial for agricultural and biomedical application. Nowadays, much attention has been paid to the secure and sustainable management of lignocellulosic wastes. Cellulose is regarded as one kind of slowly biodegradable organic components owing to its complex crystalline structure. This study investigated the effect of O_2-containing gas NBW on anaerobic digestion (AD) of cellulose for methane production. Results show that the cumulative methane yields from the reactors with Air-NBW (193 NmL/g-$VS_{reduced}$), N_2-NBW (196 NmL/g-$VS_{reduced}$) and O_2-NBW (233 NmL/g-$VS_{reduced}$) addition were increased by 8–30% in comparison to the control (with the same amount of deionized water addition) (179 NmL/g-$VS_{reduced}$). Under NBW addition, the reductions in cellulose content and cellulose crystallinity were respectively enhanced by 8–14% and 9–21% during AD, in which cellulase activity was elevated by 10–38%. The O_2-NBW reactor was found to have the highest electron transport system activity, increasing by 1.7 times compared to the control, most probably due to the collapse of O_2-nanobubbles and release of O_2 resulting in micro-oxygen environment under the test condition. Besides, microbial community analysis suggests that the direct interspecies electron transfer could be quickly established with the addition of NBW. Results from this study also shed light on the mechanisms involved in the bioconversion of cellulose to methane via AD supplemented with NBW.

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https://doi.org/10.1016/j.cej.2020.125652
Received 27 December 2019; Received in revised form 15 May 2020; Accepted 25 May 2020
Available online 06 June 2020
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1. Introduction

Fossil fuel is still the main source of global energy, contributing to both greenhouse gases emission and worse regional air quality. A robust global economic growth and increasing demand on temperature control during daily life have driven the rapid growth of global energy consumption. European Biogas Association (EBA) has developed a coherent strategy for realizing the optimal use of biogas, syngas and biomethane from anaerobic digestion (AD) and biomass gasification to face energy crisis and environmental pollution [1]. In this context, there is a great demand for seeking the clean or pollution free methods to enhance methane production from AD of various organic solid wastes.

Lignocellulose is one kind of polymeric materials with intricate three-dimensional network, in which the content of cellulose is up to 50% [2]. The secure and sustainable management of lignocellulosic biomass has also been considered as a prominent alternative for petroleum [3]. In addition, biomethane production from lignocellulosic biomass through AD process has become more attractive due to its environmental sustainability and low cost [4]. To improve AD efficiency of lignocellulose, various pretreatments have been attempted, including physical, chemical and biological methods. For instance, hydrothermal pretreatment of lignocellulosic biomass gained the highest biodegradability and biochemical methane potential [5]. Sambusiti et al. claimed that alkaline pretreatment could enhance the methane production from lignocellulose [6]. Methane production from lignocellulose can also be enhanced after pretreatment by a pure strain *Bacillus Subtilis* [7]. Although the above-mentioned methods can achieve some improvement on methane production from AD of lignocellulosic materials, the increment of methane yield is difficult to balance the possible secondary pollution and the high operation cost, mainly due to the large amount of chemicals addition and/or additional energy consumption in these pretreatments.

Most recently, the advancements in micro-oxygen technology reflect great potentials to address the low hydrolysis rate of highly recalcitrant feedstocks (like lignocellulosic biomass in this study) [8] and low metabolic activity of microbes [9]. In general, oxidation-reduction potential (ORP) value in the range of 0 to −470 mV can be defined as micro-oxygen environment that could improve methane yield and volatile fatty acids (VFAs) production to a certain extent in AD process [8,10]. Hydrolysis is the rate-limiting AD step of refractory substrates such as lignocellulosic biomass [11]. The underlying rationale of an enhanced AD process is usually relating to the augmentation of microbial diversity and activity (facultative heterotrophs) resulting in improved hydrolysis of substrate [10]. For example, the micro-aerobic AD system was detected to have a higher proportion of *Firmicutes* phylum associated with a higher substrate hydrolysis rate [8]. An intermittent micro-aeration system can also accelerate VFAs conversion by facultative bacteria and improve methane yield by hydrogenotrophic methanogens, respectively [10]. In order to achieve the micro-aeration purpose, high costs are required for electricity and maintenance of the mechanical parts when conventional mechanical aerator or diffuser systems are applied [12]. On the other hand, some chemical oxidants (i.e. H_2O_2 and calcium peroxide) can release peroxides to react with water and elevate O_2 levels, but they also kill the microbes and pollute aquatic ecosystems [13]. All these works indicate that the micro-aeration biotechnology would be more promising if it is non-toxic with no chemical addition, especially for the enhancement of AD process, in order to meet the sustainable development goals (SDGs).

Nanobubble water (NBW) contains 10^6–10^8 fine bubbles per milliliter with diameter < 1 μm [14], which can induce many intriguing properties. These nanobubbles (NBs) possess high mass transfer efficiency, high surface area, and long residence time [15]. NBs in NBW can increase the contact surface area between the liquid and solid phases. The hydrophobic attraction force property of NBs is also helpful for them to adhere onto the solid surface, thus promoting mass transfer, chemical reactions and metabolism [16]. NBW can be relatively stably stored for three months [17], thus it is likely able to maintain a long-lasting O_2 concentration if being added into AD reactors. Air-NBW was presumed to create micro-oxygen environment that greatly improved the hydrolysis-acidification stage of high cellulose loading AD [18]. O_2-NBW was trialed to control methane emission from algae bloom waters [19], which has also been found to augment the anaerobically digested sludge and then improve methane production from corn straw [9]. In addition, the mass transfer rate of O_2 to the liquid phase in O_2-NBW was claimed to be 125 times faster than that of bulk bubble (diameter > 1000 nm) [20]. Seen from the above works, O_2-containing gas NBW has great potentials for enhanced AD of lignocellulosic biomass. However, whether O_2-containing gas NBW can create a micro-oxygen environment or not remains unclear, and the underlying mechanisms for the enhanced methane production from AD of lignocellulosic biomass hasn't been addressed so far. Most importantly, up to now, no reports disclosed the exact role of O_2 released by O_2-NBW on microorganisms and their metabolism in the micro-oxygen environment, in addition to the effect of O_2 concentration in NBW on methane yield.

Therefore, the present work was designed to investigate the effects of different O_2-containing gas NBWs (Air-NBW, N_2-NBW, O_2-NBW, and artificial N_2/O_2 gas mixture with different mixing ratios) addition on AD of cellulose. The main objectives of this study were: 1) to investigate the effects of NBW on electron transport system (ETS) and enzyme activity during AD of cellulose under the addition of different NBWs; 2) to quantify the cellulose content and crystallinity reduction efficiency; 3) to verify O_2-NBW's function on biochemical methane potential (BMP) using different N_2 and O_2 gas mixture derived NBWs; and 4) to elucidate the possible mechanisms by analyzing bacterial and archaeal communities for a better understanding of methane production from cellulose under NBW addition.

2. Materials and methods

2.1. Materials

In this study, the anaerobically digested sludge as inoculum for BMP test was collected from Shimodate sewage treatment plant in Ibaraki, Japan. The digested sludge was filtered through a 1-mm sieve and stored in a refrigerator at 4 °C before being used. To remove the easily biodegradable substances contained and activate microbial activity, the digested sludge was anaerobically digested at 35 ± 1 °C for one week before BMP test. Cellulose used in this study was purchased from Sigma-Aldrich (Avicel® PH-101), with an average particle diameter of 50 μm. The physicochemical properties of the cellulose and digested sludge are shown in Table 1. The cellulose crystallinity is very high, about 80–81%, indicating that microcrystalline cellulose is very difficult to be degraded [21]. N_2 (purity ≥ 99.9995%) and O_2 (purity ≥ 99.9995%) gases being used for NBW generation were purchased from Taiyo Nippon Sanso Co., Ltd. (Japan). Room air in the

Table 1
Physicochemical characteristics of cellulose and inoculum used in this work.

Items	Unit	Inoculum	Cellulose
pH	–	6.8 ± 0.1	ND
TS	%, w.w.	1.77 ± 0.04	98.90 ± 0.2
VS, of TS	%, w.w.	72.88 ± 3.2	99.98 ± 0.3
Total carbon	%, d.w.	28.1 ± 2.1	43.5 ± 0.4
Total nitrogen	%, d.w.	2.9 ± 0.6	ND
TVFAs	mg/L	46.3 ± 1.4	ND
Soluble total organic carbon	mg/L	126.8 ± 2.3	ND
Soluble carbohydrates	mg/L	69.3 ± 1.2	ND
Soluble proteins	mg/L	8.2 ± 0.3	ND
Cellulose	%, d.w.	15.8 ± 1.2	98.6 ± 2.3
Cellulose crystallinity	%	46.80 ± 0.6	80.79 ± 2.0

d.w. – dry weight basis, ND – not detected, TS – total solids, TVFAs – total volatile fatty acids, VS – volatile solids, w.w. – wet weight basis.

laboratory was used to produce Air-NBW in this study.

2.2. Preparation of NBW

NBW was prepared as previously described [18]. Briefly, 1 L deionized water (DW) in a 5 L plastic beaker was injected into the HACK FB11-UFB generator (HACK UFB Co., Ltd., Yamanashi, Japan) with different gases to obtain 'water containing NBs', namely NBW. Gas intake speed and DW circulation flow rate of the HACK FB11-UFB generator were controlled at 0.05 L/min and 3 L/min, respectively. The NBW generation process lasted for 20 min at room temperature (25 ± 1 °C). NBW was produced in a combination of pressurized mechanical cyclic and spiral liquid flow. One liter of air, N_2, O_2 or the N_2/O_2 gas mixture at a different volume ratio of 4:1, 1:1 or 1:3 was used for NBW generation, respectively. The N_2/O_2 gas mixture was produced by using a wet gas meter (W-NK-0.5, Shinagawa Co., Japan).

2.3. BMP tests

The BMP test was used to assess the biodegradability of cellulose with NBW and DW addition. The tests were performed using 120 mL closed Schott Duran serum bottles. Each bottle, with a working volume of 70 mL, was firstly added with 35 mL of NBW or DW, then 35 mL of inoculum with biodegradable substances removed, and at last, cellulose at a concentration of 5 g/L, achieving a feed-to-microorganisms (F/M) ratio of 1 g-volatile solids $(VS)_{cellulose}$/g-$VS_{inoculum}$ with no addition of other substances like macro- and/or micro-nutrients. As reported, slightly alkaline condition can enhance the hydrolysis of cellulose [22], the limiting step of AD. In our previous studies [14,18], under the addition of NBW, VFAs accumulation usually occur during the initial stage of AD of waste activated sludge and cellulose, which may lead to an acidic environment influencing the BMP tests. Thus, in this study, after inoculation, the initial pH of the mixture was adjusted to 8.0 using 2.0 M NaOH or HCl to avoid (or alleviate) the above-mentioned VFAs inhibition effects on AD process. Then, the AD reactors were subjected to 5 min of N_2 flushing to create anaerobic conditions. All the AD experiments were conducted for 18 days in a temperature-controlled incubator at 35 ± 1 °C, and all the tests were carried out in triplicate. Six types of NBW derived from different gases (air, N_2, O_2, and the N_2/O_2 gas mixture at ratios of 4:1, 1:1 and 1:3) were tested to investigate BMP from cellulose. The reactors were labelled as Air-NBW, N_2-NBW, O_2-NBW, N_2/O_2 (4:1)-NBW, N_2/O_2 (1:1)-NBW, N_2/O_2 (1:3)-NBW, respectively. The addition of DW instead of NBW was used as control (DW reactors). The reactors without cellulose addition (only inoculum and NBW or DW) were conducted as blank to eliminate the methane amount produced by the residual biodegradable organics in the inoculum under NBW or DW addition. The methane yield from each blank was also deducted from each treatment. The cumulative methane production was calculated based on the reduced VS of cellulose. Net methane yield (NmL/g-$VS_{reduced}$) from cellulose at all conditions was calculated as previously described [18]. The BMP in this study was expressed as the volume of methane under standard conditions (1013 hPa, 273 K) as described elsewhere [9].

2.4. Analytical methods

2.4.1. General parameters

The size and number densities of NBs were measured using nanoparticle tracking analysis method by NanoSight-LM10 (Malvern Panalytical Co. Ltd., UK). Zeta potential of NBW was measured using Zetasizer Nano S (Malvern Panalytical Co. Ltd., UK). For each detection, three samples were prepared in parallel in 10 mm NMR tubes. Total solids (TS) content was analyzed by drying the samples at 105 ± 1 °C till constant weight. VS content was recorded as the loss of dry mass after burning at 600 ± 2 °C for 3 h. pH was determined using a semi-solid pH meter (Testo 206, Germany). Dissolved oxygen (DO) was determined by a HQ40d Portable Meter (Hach, USA). ORP was measured using a YK-23RP ORP meter (Lutron electronic enterprise Co., Ltd., Taiwan). Conductivity was determined using an AS710 conductivity meter (Mechworth Co. Ltd., Thailand). Soluble proteins and carbohydrates were quantified using Coomassie Brilliant Blue (CBB) and phenol–sulfuric method with bovine serum albumin (BSA) and L-glucose as standard, respectively. The volume of biogas production from batch AD experiment was measured using glass syringe (20 mL) method, and the bottle was manually shaken for 2 min before measurement. Methane content was analyzed by a gas chromatograph (GC-8A, Shimadzu, Japan) packed with a thermal conductivity detector (80 °C) and a Porapak Q column (60 °C). The reaction residue was centrifuged at 10,000 rpm for 30 min to obtain the supernatant. After being filtered through a 0.22 μm microporous membrane, the soluble total organic carbon (STOC) was determined by a total organic carbon analyzer (TOC-V_{CSN} with ASI-V autosampler, Shimadzu). For the measurement of VFAs concentration, phosphoric acid solution (3%) was added to the filtrate to acidify to pH around 4.0 before being analyzed by gas chromatograph (GC-8A, Shimadzu) packed with Unisole F-200 30/60 column and flame ionization detector (FID). The pressure of carrier gas (N_2) was maintained at 200 kPa. The temperatures of column and injector were 150 °C and 180 °C, respectively. As for the solid total organic carbon (TOC) analysis, the reaction residue was dried at 60 ± 2 °C for 72 h and grounded into powder, and then analyzed by the solid sample combustion unit (TOC-V_{CSN} with SSM-5000A, Shimadzu). Total nitrogen (TN) concentration was measured with UV spectrophotometric method after samples being digested by alkaline potassium persulfate [23].

2.4.2. Theoretical methane production and biodegradability

Theoretical methane yield (TMY) of cellulose was estimated according to its elemental composition C, H and O by using the following Eqs. (1) and (2) [24].

$$C_aH_bO_c + \left(\frac{4a - b - 2c}{4}\right)H_2O \rightarrow \left(\frac{4a + b - 2c}{8}\right)CH_4 + \left(\frac{4a - b + 2c}{8}\right)CO_2 \quad (1)$$

$$TMY = 22.4 \times \frac{((4a + b - 2c)/8)}{12a + b + 16c} \quad (2)$$

where a, b and c represent the molar fractions of C, H and O in cellulose, respectively.

Anaerobic biodegradability (BD) of each sample was calculated from the values of experimental methane yield (EMY) and TMY according to Eq. (3) [25].

$$BD\ (\%) = \frac{EMY}{TMY} \times 100 \quad (3)$$

The modified Gompertz model was also used to analyze the methane production rate during AD [26]. The model as in Eq. (4) is characterized by an S-shaped curve with a clear inflection point.

$$M = M_0 \exp\left\{-\exp\left[\frac{\mu_m e}{M_0}(\lambda - t) + 1\right]\right\} \quad (4)$$

where M refers to the simulated cumulative methane yield (CMY) (NmL/g-$VS_{reduced}$); M_0 represents the simulated maximum CMY (NmL/g-$VS_{reduced}$); μ_m represents the maximum methane production rate (NmL/(g-$VS_{reduced}$·d)); e is the Euler's number, 2.718; λ denotes the lag phase time (days); and t stands for the digestion duration (days).

2.4.3. Cellulose content, cellulase activity and crystallinity index (CrI)

The cellulose content was determined according to the method as previously described [18]. Since it is very difficult to separate the 50 μm cellulose particles from the digested sludge, the cellulose content during the AD was estimated by subtracting the cellulose content of the blank group (the inoculum only) from the total cellulose content. The

cellulose reduction during AD process was calculated by Eq. (5)

$$\text{Cellulose reduction } (\%) = \frac{[C]_{initial} - [C]_{final}}{[C]_{initial}} \times 100 \quad (5)$$

where $[C]_{initial}$ and $[C]_{final}$ are the cellulose contents in the samples before and after AD process, respectively.

Cellulase activity was analyzed as described by Hua et al. [27]. Briefly, after centrifugation of the sample at 10,000 rpm for 10 min, 0.6 mL of the supernatant was taken and mixed with 0.6 mL of 1% (w/v) carboxymethyl cellulose (CMC) solution, and then incubated at 60 ± 2 °C for 10 min. Later, 1.8 mL of 3,5-dinitrosalicylic acid (DNS) reagent (1% NaOH, 20% Rochelle salt, 2% phenol, 0.005% Na_2SO_3 and 1% 3,5-dinitrosalicylic acid) was added into the mixture, which was then incubated at 100 °C for 10 min. After that, the mixture was cooled down in an ice bath for 5 min, and the absorbance was measured at 540 nm. In this study, one unit of cellulase activity (U/(mL·min)) was defined as the amount of enzyme releasing 1 mmol of reducing sugar per minute under the above conditions.

Crystallinity index (CrI) of cellulose in each sample before and after batch AD experiment was measured by the wide-angle X-ray diffraction (D8 Advance, BrukerAXS, Germany). This machine was operated at 40 kV and 40 mA with an angular range from −100 to 168. The samples were scanned at 2θ range of 10° to 40° at a scanning speed of 1.0°/min. The CrI was calculated according to Eq. (6) [28].

$$\text{Crystallinity index } (\text{CrI},\%) = \frac{[I]_{crys} - [I]_{amor}}{[I]_{crys}} \times 100 \quad (6)$$

where $[I]_{crys}$ is the intensity for crystalline portion of biomass at 2θ = 23°, and $[I]_{amor}$ is the intensity for amorphous portion of biomass at 2θ = 19°, respectively.

2.4.4. Analysis of electron transport

ETS activity of the methanogenic communities was measured using the INT (2-(p-iodophenyl)-3-(pnitrophenyl)-5-phenyltetrazolium chloride) reduction method according to Zhang et al. [29]. Firstly, 0.5 mL sample was taken from each reactor and added into a 10 mL centrifuge tube. After addition of 0.1 mL 0.2% INT solution, the tube was incubated under dark condition and shaken in a water bath oscillator at 100 rpm and 37 ± 1 °C for 30 min. Then, 1 mL methanol (37%) was added to terminate the reaction. The supernatant was discarded after centrifugation at 10,000 rpm and 4 °C for 5 min. Afterward, 5 mL methanol was added into the sediment and the sample was again shaken in dark in the water bath oscillator (100 rpm and 37 ± 1 °C) for 10 min. The supernatant obtained after centrifugation (10,000 rpm, 5 min) was measured at 485 nm by a spectrophotometer. The sediment was dried at 105 ± 1 °C to a constant weight and weighed. ETS activity was calculated according to Eq. (7).

$$U = \frac{A_{485} \times V}{K_T \times W \times t} \quad (7)$$

where U is ETS activity $(mg\cdot(g\cdot h)^{-1})$; A_{485} is absorbance at 485 nm; V is the volume of methanol (mL); K_T is the slope of calibration curve; W is the dry weight of the sediment (mg); and t is reaction time (0.5 h).

2.4.5. Analysis of bacterial and archaeal communities

The sludge samples were collected at the end of experiment for bacterial and archaeal community analysis, respectively. Microorganisms in the digestate were identified by 16S rRNA gene sequencing. Microbial DNA was extracted from sludge samples using the E.Z.N.A.® soil DNA Kit (Omega Bio-tek, Norcross, GA, US) according to manufacturer's protocol. Polymerase chain reaction (PCR) was conducted using a Gene Amp 9700 (ABI, USA). For the analysis of total bacteria, DNA samples were amplified with primers 338F (5′-ACTCCT ACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGTWTCT-AAT-3′) by the thermocycler PCR system. For the identification of the archaea, 109F (5′-ACTGCTCAGTAACACGT-3) and 519R (5′-CGTATTA CCGCGG CTGCTGG-3′) were used as the primer pair for amplification. PCRs were performed with 10 ng template DNA in 20 μl volumes. Each reaction contained 5 × FastPfu Buffer 4 μl, 2.5 mM dNTPs 2 μl, Forward Primer (5 μM) 0.8 μl, reverse Primer (5 μM) 0.8 μl, FastPfu Polymerase 0.4 μl, BSA 0.2 μl. The following cycle parameters were applied during the operation: initial denaturation at 95 °C for 3 min; 27 cycles at 95 °C for 30 s, at 55 °C for 30 s, and at 72 °C for 45 s; and the final extension at 72 °C for 10 min. The resultant PCR product was extracted from a 2% agarose gel and further purified using AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA, USA), quantitatively using QuantiFluorTM-ST (Promega, USA) according to the manufacturer's protocol. High-throughput sequencing was performed using an Illumina MiSeq platform (Illumina Lnc., CA, USA) in Shanghai Majorbio Bio-Pharm Technology Co., Ltd. (Shanghai, China). Phylum and genus levels were used to analyze the diversity of bacterial and archaeal sequences, respectively. Difference in community composition among treatments was also investigated by principal component analysis (PCA).

2.5. Data processing and statistical analysis

All data were calculated using Microsoft Office Excel 2018. One-way analysis of variance (ANOVA) was conducted to analyze the statistical difference among the NBW and DW tests, and significant difference was assumed at $p \leq 0.05$. All figures in this study were plotted using Origin 2018.

3. Results and discussion

3.1. Micro-oxygen environment created by O_2-NBW

Micro-oxygen environment has been claimed to have enhancement effect on methane yield from AD of lignocellulosic biomass [30]. In this regard, the oxygen contained in NBs is difficult to be detected by the standard DO meter or Winkler oxygen titration method [15], although the DO value of O_2-NBW was slightly higher than Air-NBW and N_2-NBW after reaching stable (Fig. 1a). Since ORP has a linear relationship with the logarithm of oxygen concentration, and a DO of 0.1 mg/L is equivalent to an ORP of −96 mV in the aqueous phase [10], ORP value in the range of 0 to −470 mV is defined as micro-oxygen environment [8,10]. In this study, the highest ORP value was detected in the O_2-NBW reactor (−412 ± 14 mV), followed by the Air-NBW (−475 ± 1.4 mV), N_2-NBW (−480 ± 5.7 mV) and DW (−489 ± 1.4 mV) reactors under the test conditions (Fig. 1b). There is no significant difference ($p = 0.2 > 0.05$) in ORP between the Air-NBW and N_2-NBW reactors, likely due to the O_2 content in the laboratory air is not high enough to create a micro-oxygen environment during NBW generation. In addition, in the O_2-NBW reactor the oxygen content was calculated as 4 μl higher than that in the DW reactor. As reported, the optimal ORP for methanogens is lower than −350 mV [31], indicating that the micro-oxygen environment brought about by O_2-NBW addition may not inhibit the AD of cellulose to produce methane. The observations on DO and ORP changes in this study are similar with the O_2-NBW applications in the remediation of anaerobic environment by Wang et al. [32], in which O_2-NBW was found to prolong the survival time of aerobic zebrafish due to the O_2 released from O_2-NB. Results from this study indicate that O_2-NBW may provide a micro-oxygen environment that would enhance methane production from AD of lignocellulosic biomass.

3.2. Changes in TOC removal and cellulose content in BMP tests

TOC was measured to determine the effect of NBW on the reduction of organic matter during AD. Fig. 2a shows the initial and final TOC concentrations and TOC removals during AD. After 18 days of AD, TOC

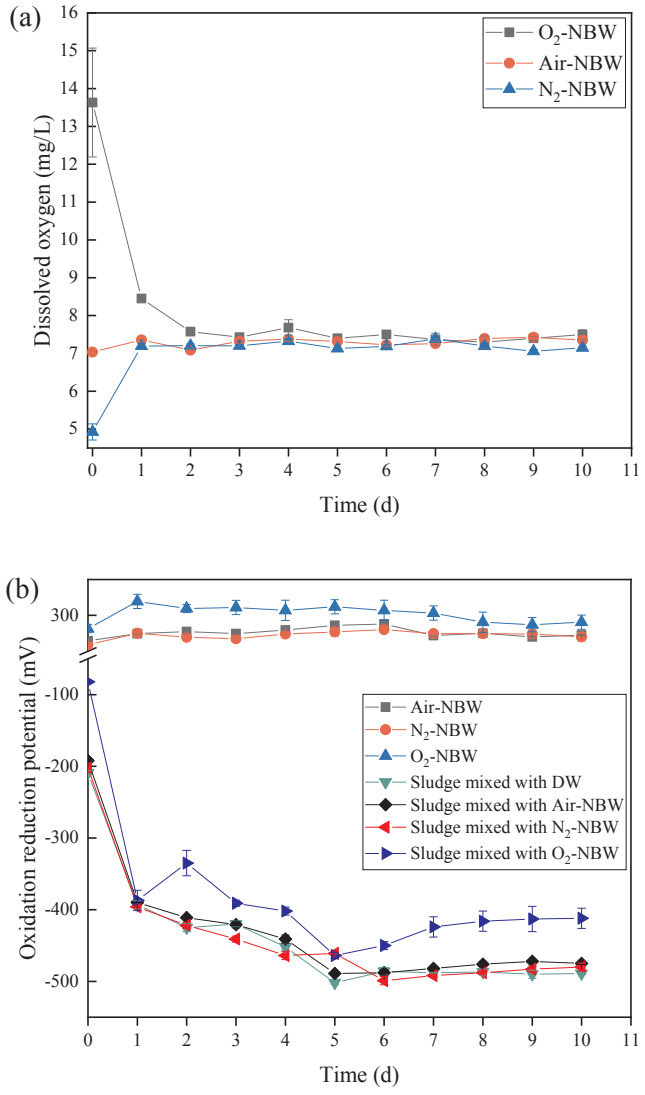


Fig. 1. Changes in dissolved oxygen (a) and oxidation reduction potential (b) during 10 days' storage at 35 ± 1 °C. NBW- nanobubble water, DW- deionized water.

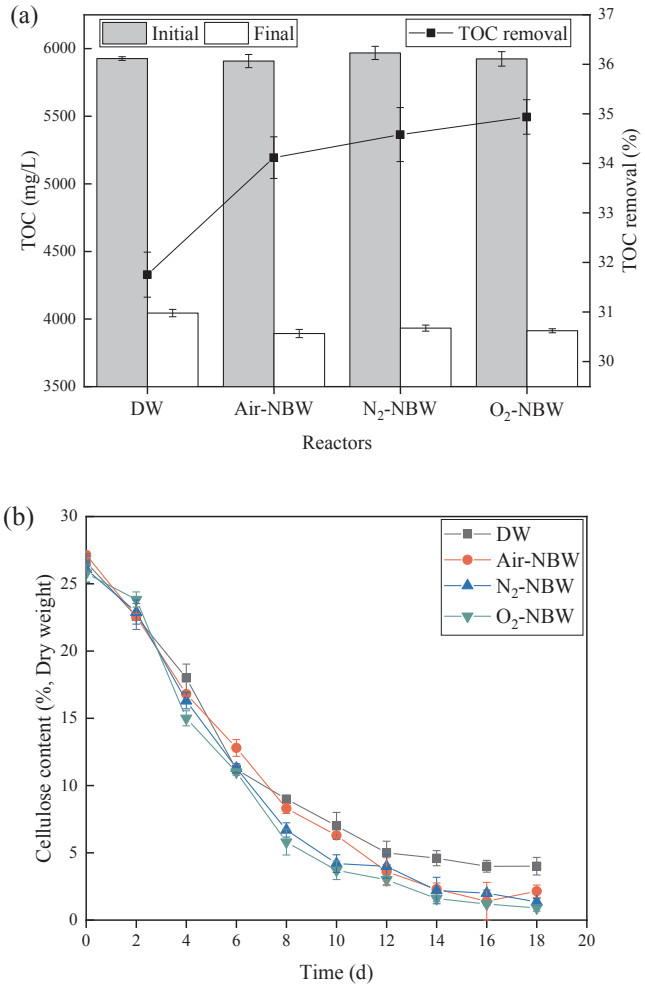


Fig. 2. Changes in total organic carbon (TOC) of the initial and finial samples and TOC removals (a), and reductions in cellulose content (b) during anaerobic digestion of cellulose under deionized water (DW) and different nanobubble water (NBW) addition.

was significantly reduced in all the reactors. The average TOC removal reached to 31.8%, 34.1%, 34.6%, and 34.9% in the DW, Air-NBW, N_2-NBW and O_2-NBW reactors, respectively. The results indicated that NBW addition enhanced the utilization efficiency of cellulosic compounds, with the highest TOC removal achieved in the O_2-NBW reactor, increasing by 9.7% ($p = 0.03 < 0.05$) compared to that of the DW reactor. Cellulose reduction was detected in all the reactors (Fig. 2b). As shown, the cellulose content gradually decreased along with the 18 days' AD. In the O_2-NBW reactor, cellulose reduction reached to 96.5%, about 13.6% higher compared to the DW reactor (84.9%), followed by N_2-NBW (94.8%) and Air-NBW (92.1%) reactors, respectively. The hydrolysis of cellulose particles can form an oligosaccharide layer, which may hinder the further degradation of cellulose particles as deduced by Vaquerizo et al. [21] who did not mention how to degrade the oligosaccharide layer. The addition of NBW may enhance the decomposition of the oligosaccharide layer due to the high mass transfer efficiency and miniaturization of NBs as well as high mobility of water molecules in NBW [18]. Thus, NBW could increase availability of small molecules to methanogens, resulting in the enhanced cellulose degradation by microbes during AD (Fig. 2b and Fig. 6). This observation indicates that NBW could enhance the rate-limiting step, hydrolysis of cellulose during AD. Restated, the degradation of cellulose mainly depends on cellulase activity which might be also enhanced after NBW addition.

3.3. Cellulase activity in BMP tests

Cellulase activity can be used to indicate the crystalline cellulose degradation ability of microbial consortia. It can break down the β-(1,4)-glycosidic linkages to degrade cellulose, with the concomitant release of glucose monomers [33]. In this study, cellulase was analyzed as a representative enzyme involved in the hydrolysis of crystalline cellulose. As shown in Fig. 3a, the cellulase activity was relatively low during the initial two days, which was apparently enhanced from day 2 to day 4, and then decreased from day 6 to day 12. This result is to some extent similar with the change of cellulase activity in the AD of corn straw and cow dung [34]. The highest cellulase activity (0.246 U/(mL·min)) was detected in the Air-NBW reactor, about 37.9% higher than that in the DW reactor, which maintained for 2–3 more days compared with other reactors. While during the last stage (12–18 days), the N_2-NBW reactor exhibited the highest cellulase activity (0.212 U/(mL·min)), about 12.6% higher than that in the DW reactor. In addition, the cellulase activity in the Air-NBW and O_2-NBW reactors showed no significant difference ($p = 0.3 \gg 0.05$) on day 18, which were 10.1% higher than that in the DW reactor. These results suggest that NBW addition could enhance the cellulase activity, thus promoting crystalline cellulose hydrolysis during AD. However, these results cannot

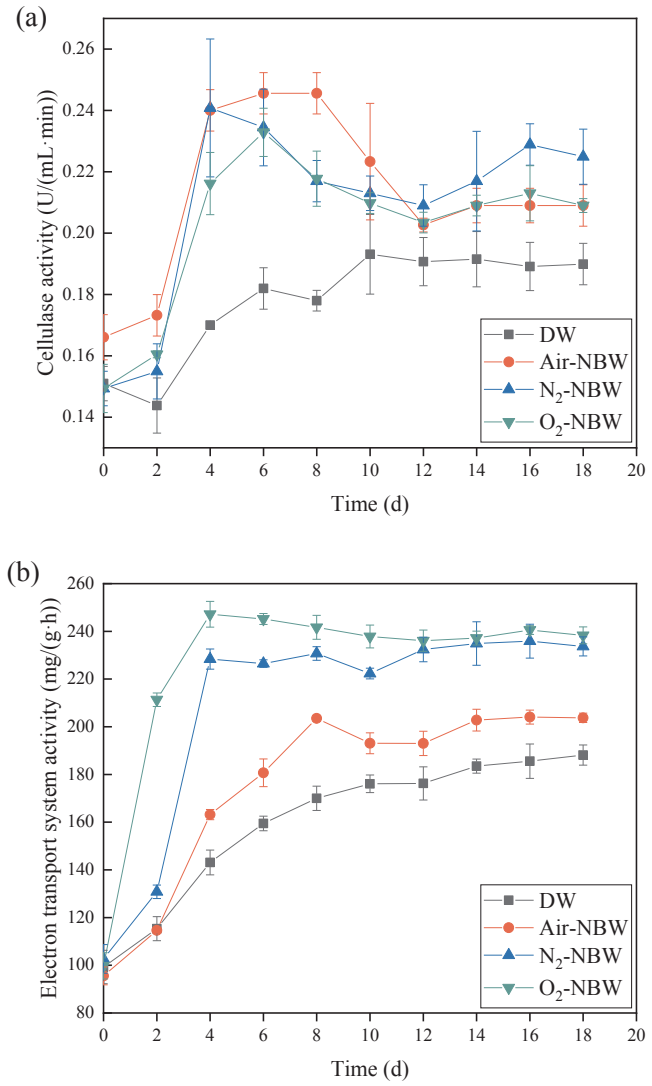


Fig. 3. Changes in cellulase activity (a) and electron transport system activity (b) during 18 days' AD under DW, Air-NBW, N_2-NBW and O_2-NBW NBW addition. DW- deionized water, NBW- nanobubble water.

explain the highest methane production from the O_2-NBW reactor. The AD of cellulose basically follows four stages, i.e. hydrolysis, acidification, acetogenesis and methanogenesis. Cellulose hydrolysis only occurs during the hydrolysis stage of AD. Therefore, it could be preliminarily inferred that the methane yield was affected by other three stages with NBW addition. More specifically, the trace O_2 concentration in the O_2-NBW reactor may exert significant effect on methane production.

To further clarify the crystalline cellulose hydrolysis during AD, Table 2 lists the cellulose crystallinity reduction in the test reactors. The highest reduction in cellulose crystallinity (38.6%) was detected in the O_2-NBW reactor, about 21.0% increase compared to the DW reactor (31.9%). The cellulose crystallinity reduction was enhanced by 8.4–17.5% under other NBW additions. Reduction in cellulose crystallinity can increase its surface area and accessibility for better enzymatic hydrolysis/microbial degradation. According to the observations by Wang et al. [18], NBW addition can enhance the reduction of cellulose crystallinity, probably due to its improvement effects on the metabolic activity of microorganisms [18]. The microbial metabolic activities after NBW addition were discussed in the ETS tests in sections 3.4.

3.4. INT-electron transport system activities in BMP tests

The microbial bioactivity during the 18 days' AD with NBW addition were assessed by monitoring the electron transport rate during the microbial metabolism. It is correlated with intracellular ETS through measuring the rate of INT reducing 2-(p-iodophenyl)-3-(p-nitrophenyl)-5-phenyl tetrazolium chloride formazan (INTF) by providing a super electron donor (NADH and succinic acid) [29]. In this study, as shown in Fig. 3b, initially the ETS activity increased rapidly in all the reactors, which was consistent with the cellulase activity changes in these AD systems. The highest ETS activity was detected in the O_2-NBW reactor on day 4, reaching 247.2 mg/(g·h), which was approximately 1.7 times that of the DW reactor, followed by N_2-NBW (1.6 times) and Air-NBW (1.1 times) reactors (Fig. 3b). It has been confirmed that the ETS activity is closely associated with the methanogenic efficiency (Fig. 5a). ETS generally occurs in the cell membrane of bacteria [35]. It can transfer electrons from the coenzymes produced by the Szent-Györgyi-Krebs cycle (NADH, NADPH) to the terminal electron acceptor in the oxidative phosphorylation (electron transport) pathway. The surface of NBs has negative charges [16], which could attract NAD^+; NBs also have higher speed of motion, favoring the movement of NAD^+ that is used for reduction reaction to generate NADH. This might be the reason why NBW addition promoted the ETS in the reactors (Fig. 6). Ahmed et al. found that NBW could contribute to seed germination and plant growth [36], possibly due to the promotion effect of NBs on ETS activity.

O_2 can increase ETS activity because O_2 may easily act as an electron acceptor in ETS [35]. A significant amount of energy can be released by the reduction of O_2 with NADH, which then produces a proton gradient that drives the synthesis of ATP and the transport of other metabolites (Fig. 6). These deductions associated with ETS changes also suggest that the O_2-NBW reactor may have higher potential for creating micro-oxygen environment than other reactors, which might be one major reason for the more enhanced methane production from the O_2-NBW reactor. Still, future research is demanding on more direct evidence for synthesis of ATP and the transport of other metabolites under NBW addition condition. The positive effect of NBW addition on ETS activity indicates that NBW could increase the bioactivity of microbes by accelerating the electron transport, and then contribute to the enriched microbial community.

3.5. Microbial community analysis

3.5.1. Bacterial phyla

Fig. 4a shows the relative abundance (RA) of bacterial community at phylum level in the DW, Air-NBW, N_2-NBW, O_2-NBW, N_2/O_2 (4:1)-NBW, N_2/O_2 (1:1)-NBW, and N_2/O_2 (1:3)-NBW reactors. The RA of *Thermotoga*, *Chloroflexi*, *Bacteroidetes*, *Spirochaetae*, *Synergistetes* and *Firmicutes* in the NBW reactors were higher than those in the DW reactor. Most of them are known phyla of hydrolytic and acidogenetic bacteria. *Thermotogae* possesses high capability of degrading lignocellulosic biomass by producing hydrolytic enzymes [37]. *Chloroflexi* may accelerate the cellulose hydrolysis as it is regarded as a multicellular filamentous microorganism that hydrolyzes carbohydrates [38]. The abundant bacterial phylum in all the reactors was *Bacteroidetes* that are responsible for hydrolysis of lignocellulosic biomass to produce acetate, butyrate and propionaye [39]. The RA of *Bacteroidetes* is the highest in the O_2-NBW reactor, which may contribute to the increased efficiency on cellulose degradation. *Spirochaetae* belongs to acetogens, and is capable of acetogenesis, which may accelerate the transfer of VFAs to acetate [40]. *Synergistetes* can reduce the accumulation of propionic acid and butyric acid due to its efficient syntrophic oxidation capability [41], which may contribute to the relatively stable pH during the whole AD (Fig. S1). *Firmicutes* can produce extracellular enzymes such as protease, lipase and cellulase, and then involve in the degradation of organic compounds and the formation of VFAs [40]. The

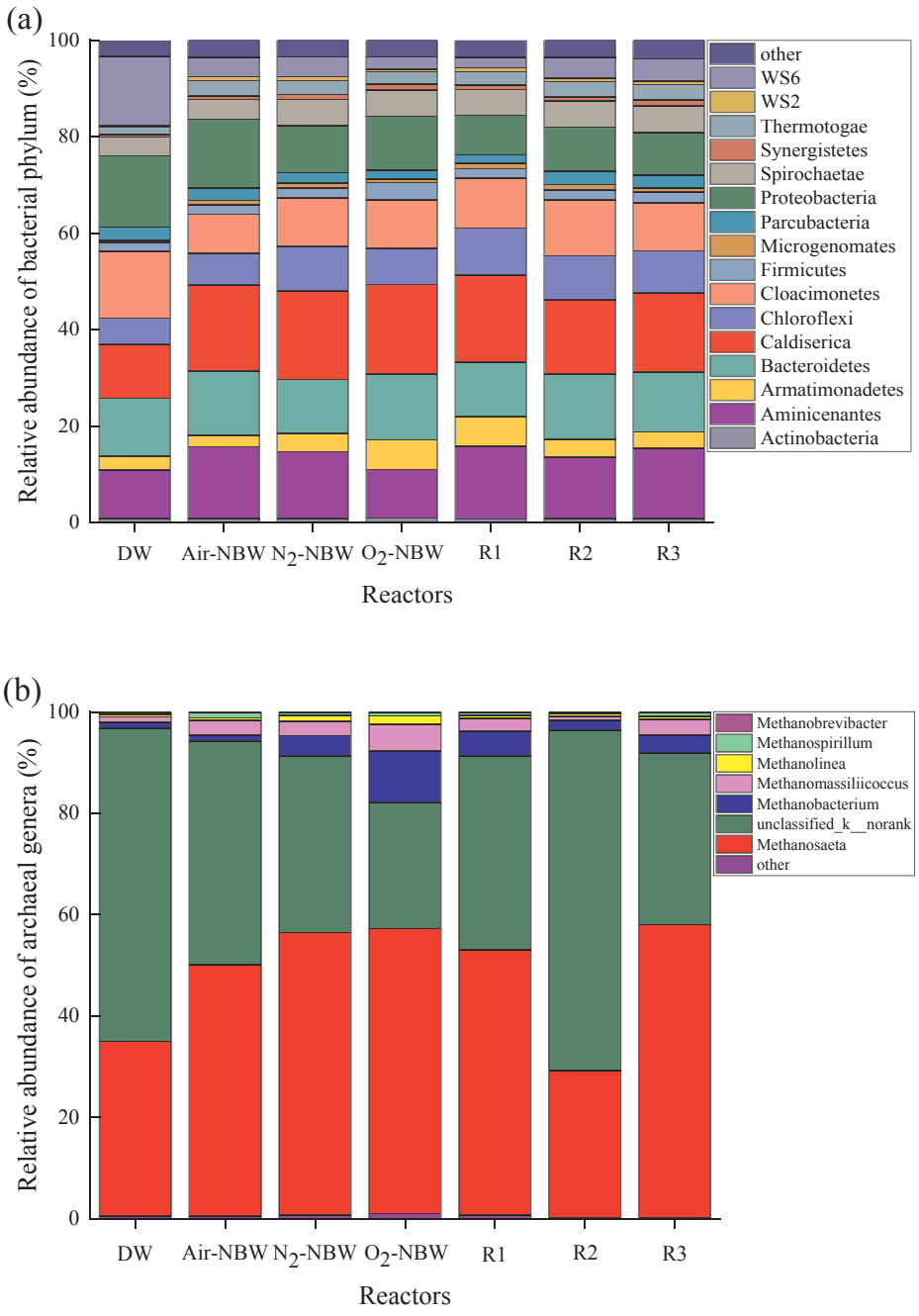


Fig. 4. The relative abundance of prevalent bacterial phylum (a) and detected archaeal genera (b). DW- deionized water, NBW- nanobubble water. N_2/O_2 (4:1)-NBW; N_2/O_2 (1:1)-NBW; N_2/O_2 (1:3)-NBW are denoted as R1, R2 and R3, respectively.

results from Fig. 3a can explain the increased activity of cellulase in the NBW reactors. As shown in Fig. 4a, the RA of *Firmicutes* reached the highest in the O_2-NBW (3.6%) reactor, followed by the N_2/O_2 (1:3)-NBW (2.2%), N_2/O_2 (1:1)-NBW (2.1%), Air-NBW (2.0%), N_2/O_2 (4:1)-NBW (1.9%) and DW (1.8%) reactors, reflecting a decrease trend as the oxygen content decreased. The above observation to some extent agrees with Fu et al. who found that the relative abundance of *Firmicutes* after micro-aeration treatment was 6% higher than that of the control group [30]. As expected, bacterial phyla analysis suggested that the micro-oxygen environment created by O_2-NBW addition might promote the enrichment of hydrolytic bacteria in the AD reactors.

3.5.2. Archaeal genus

The RA of archaeal community at the genus level is shown in Fig. 4b. The methanogenic archaea were dominant, accounting for 37.4%, 55.1%, 64.6%, 70.7%, 60.7%, 34.8% and 65.5% of archaeal 16S rRNA gene sequences recovered from the sludge samples in the DW, Air-NBW, N_2-NBW, O_2-NBW, N_2/O_2 (4:1)-NBW, N_2/O_2 (1:1)-NBW, and N_2/O_2 (1:3)-NBW reactors, respectively. As seen, *Methanobacterium*, which is highly related to hydrogenotrophic methanogenesis and can enhance interspecies H_2 transfer during AD [42], was enriched in all the NBW reactors, especially in the O_2-NBW reactor. The RA of *Methanobacterium* was 10.1% in the O_2-NBW reactor, while it was only 1.2% in the DW reactor. Fu et al. also noticed that the RA of *Mathanobacterium* was doubled under microaerobic condition [30]. These observations further imply the function of micro-oxygen condition provided by O_2-NBW addition on AD process.

In the Air-NBW and O_2-NBW reactors, the RA of

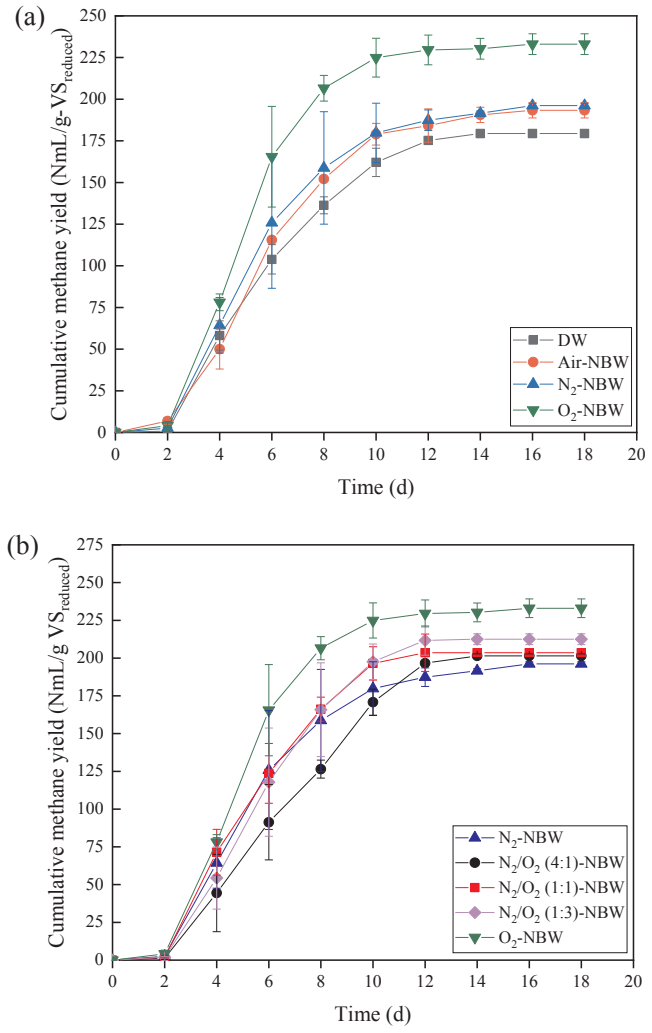


Fig. 5. Cumulative methane yields from the DW, Air-NBW, N_2-NBW and O_2-NBW reactors (a), and the different ratios of N_2/O_2 gas mixture-NBW reactors (b) during 18 days' AD of cellulose. DW- deionized water, NBW- nanobubble water.

Methanomassiliicoccus was relatively high, about 2.9% and 5.3%, respectively in comparison to 1.0% in the DW reactor. *Methanomassiliicoccus* can utilize H_2 and methanol or methylamines to produce methane [43]. A higher RA of *Methanospirillum* was detected in the NBW reactors than that in the DW reactor. *Methanospirillum* is considered as the major contributor to methane production from cellulose [44]. Besides, the RA of *Methanolinea* was higher in the O_2-NBW and N_2-NBW reactors, which can use H_2 for growth and methane production [45]. These results might explain why almost no H_2 was detected throughout the whole AD process. In addition, interspecies H_2 transfer can contribute to the reduction of pressure-H_2, which may enhance the hydrolysis of organic matter resulting in improved methane production [46]. A high pressure-H_2 may have a persistent inhibition effect on the hydrolytic and acetogenic microbial activities, which then inhibits the production or secretion of extracellular enzymes by retroinhibition, reducing the physiological activity of the microorganisms [47].

The genus level identification showed that *Methanosaeta* was the dominant genus, which is a strict aceticlastic methanogen genus and can produce methane using acetic acid and CO_2 through direct electron transfer (DIET) via directly accepting electrons from *Geobacter* [48,49]. *Methanosaeta* usually plays an important role in the methanogenic process, contributing to about 70% of methane produced. Fig. S2 and Fig. S3 demonstrate that NBW has a higher zeta potential and a higher conductivity, especially the O_2-NBW. Li et al. proposed that the surface conductivity of particles may be responsible for the zeta potential [50]. So, it is inferred that the high conductivity of O_2-NBW might be attributable to its high zeta potential. Conductive materials can strengthen the DIET in the complex organic waste [51]. Therefore, the increase in RA of *Methanosaeta* suggests the potential increase of DIET after NBW addition.

In addition, a principal component analysis (PCA) was conducted based on the diversity of microbial consortia as shown in Fig. S4. As shown, PC1, displayed along the X-axis, accounted for 92.6% variance, whereas PC2, displayed along the Y-axis, amounted to 6.4% variance. The PCA ordinations showed that bacterial and archaeal communities were significantly different between the DW and NBW reactors. This observation reveals that the addition of NBW could enrich the related species, which probably promoted the AD performance of cellulosic biomass.

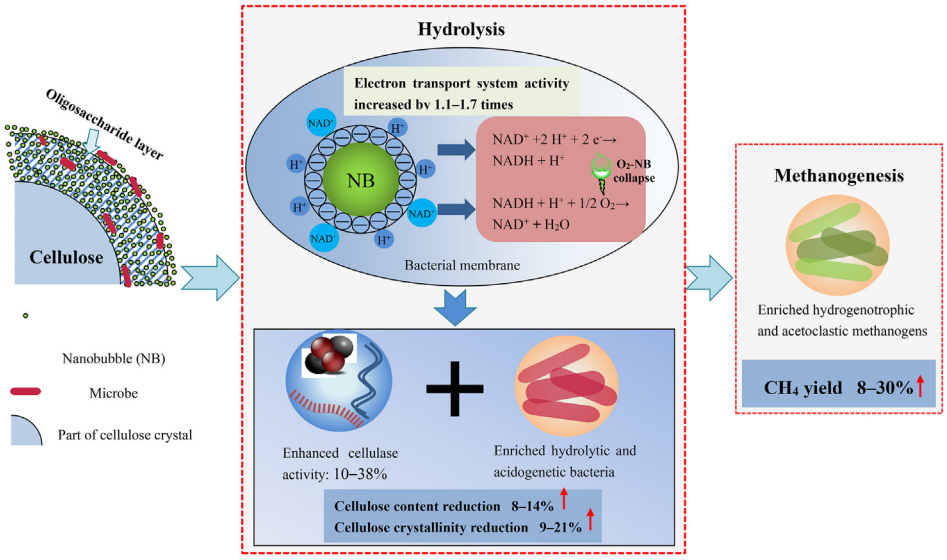


Fig. 6. Schematic pathways involved in the anaerobic digestion of cellulose under nanobubble water (NBW) addition.

Table 2
Main parameters relating to the modified Gompertz model and cellulose crystallinity.

Reactors	Cellulose crystallinity reduction (%)	TMY (NmL/g-$VS_{reduced}$)	EMY (NmL/g-$VS_{reduced}$)	BD (%)	Modified Gompertz Model			
					M_0	μ_m	λ	R^2
DW	31.9 ± 0.6 ^{e}	414.8	179.4 ± 2.4 ^{e}	43.2	181.2 ± 3.1	27.3 ± 1.9	2.1	0.996
Air-NBW	36.0 ± 0.5 ^{bc}		193.3 ± 4.5 ^{d}	46.6	193.3 ± 1.2	32.1 ± 1.2	2.4	0.999
N_2-NBW	37.5 ± 0.2^{b}		196.1 ± 1.4 ^{cd}	47.3	194.0 ± 2.1	33.8 ± 1.8	2.2	0.998
O_2-NBW	38.6 ± 0.4 ^{a}		233.0 ± 6.2 ^{a}	56.2	232.1 ± 0.7	50.0 ± 1.0	2.5	0.999
N_2/O_2 (4:1)-NBW	34.6 ± 0.8 ^{d}		201.5 ± 1.3^{c}	48.6	209.4 ± 5.0	26.4 ± 2.0	2.6	0.995
N_2/O_2 (1:1)-NBW	37.3 ± 0.6^{b}		203.6 ± 2.4 ^{bc}	49.1	206.1 ± 3.2	34.8 ± 2.6	2.2	0.996
N_2/O_2 (1:3)-NBW	35.0 ± 0.4^{c}		212.5 ± 3.6^{b}	51.2	215.7 ± 2.0	35.4 ± 1.4	2.6	0.998

BD – biodegradability, DW – deionized water, EMY – experimental methane yield, μ_m – maximum methane production rate (NmL/g-$VS_{reduced}$/day), M_0-simulated maximum cumulative methane yield (NmL/g-$VS_{reduced}$), NBW – nanobubble water, R^2-correlation coefficient, TMY – theoretical methane yield, λ – lag phase time (day). Data are expressed as mean ± SD, and the data with different superscript letters are significantly different at $p < 0.05$ according to statistical analysis.

3.6. Methane production

3.6.1. Methane yield from AD of cellulose under NBW addition

The daily methane productions from the DW, Air-NBW, N_2-NBW, O_2-NBW, N_2/O_2 (4:1)-NBW, N_2/O_2 (1:1)-NBW and N_2/O_2 (1:3)-NBW reactors are shown in Fig. S5. The daily methane production peak appeared on day 4 in the DW, N_2-NBW and N_2/O_2 (1:1)-NBW reactors, while it was on day 6 in the Air-NBW, O_2-NBW, N_2/O_2 (4:1)-NBW and N_2/O_2 (1:3)-NBW reactors; then their daily methane productions decreased. After day 16, the daily methane productions gradually decreased to 0 in all the reactors.

Fig. 5a shows that the cumulative methane yields from cellulose in the DW, Air-NBW, N_2-NBW and O_2-NBW reactors varied during the 18 days' batch BMP tests. In the first two days, little methane production was detected from the NBW addition reactors, which is similar with that of the DW reactor. After that, their methane productions increased remarkably until day 14 when their cumulative methane yields became level off. During the 18 days' BMP, the cumulative methane yields from cellulose in the O_2-NBW, N_2-NBW and Air-NBW reactors reached 233.0 ± 6.2, 196.1 ± 1.4, and 193.3 ± 4.5 NmL/g-$VS_{reduced}$, increasing by 29.9%, 9.3% and 7.7% compared with that from the DW reactor (179.4 ± 2.4 NmL/g-$VS_{reduced}$). In this study, the highest cumulative methane yield was achieved from the O_2-NBW reactor, most probably attributable to the enriched microbial community (as noted in 3.5), thus enhanced decomposition of cellulose. To clarify the effect of O_2 content in NBW on BMP, the N_2 and O_2 gas mixture NBW was examined through the BMP of cellulose at different mixing ratios (Fig. 5b). As expected, the highest methane yield was achieved from the N_2/O_2 (1:3)-NBW reactor, followed by N_2/O_2 (1:1)-NBW and N_2/O_2 (4:1)-NBW reactors. Results show that the NBW with the highest O_2 content (O_2-NBW) performed best in the methane production from AD of cellulose (Fig. 5b).

3.6.2. Analysis based on the modified Gompertz model

The simulated maximum cumulative methane yields (M_0) are very similar to the experimental methane yields, indicating that the AD of cellulose can be perfectly fitted by the modified Gompertz equation (Fig. S6), with a high correlation coefficient (R^2) ranging from 0.995 to 0.999. Table 2 also lists the model-based best-fit evaluation of kinetic parameters. The maximum methane production rate (μ_m) increased in the reactors with NBW addition compared with that in the DW reactor, indicating their enhanced AD efficiency. However, AD with NBW addition showed a little bit longer lag phase time (λ) than the DW reactor, possibly due to that the anaerobic microorganisms need a short adaption period to accommodate the new environment created in the reactors by NBW addition. In this table, the theoretical methane yield (TMY) was calculated based on the chemical formula of cellulose ($(C_6H_{10}O_5)_n$); and BD represents the biodegradability of cellulose during the AD process, which was calculated by the ratio of EMY to TMY. The BD value can be also used to predict the overall enhancement on energy recovery and to better assess the degradation of organic matter during AD process. A maximum BD value of 56.2% was achieved in the O_2-NBW reactor, followed by N_2/O_2 (1:3)-NBW (51.2%), N_2/O_2 (1:1)-NBW (49.1%), N_2/O_2 (4:1)-NBW (48.6%), N_2-NBW (47.3%), Air-NBW (46.6%) and DW (43.2%) reactors (Table 2), respectively. The results reflect that methane production from cellulose was enhanced with the addition of NBW.

3.7. Preliminary analysis on mechanisms

The pathways involved in the application of NBW for enhanced methane production from cellulose are illustrated in Fig. 6. The oligosaccharide layer produced during the first-stage of AD, namely, hydrolysis of cellulose particles exhibits a low diffusion efficiency. The cellulose particles are surrounded in a layer form, resulting in mass transfer resistance and hindered decomposition of cellulose particles [21]. In this study, the hydrophilic and negatively charged surfaces of O_2-NBs, Air-NBs or N_2-NBs can attract H^+ and NAD^+, then help to attack the oligosaccharide layer by cellulolytic and microorganisms together with cellulase due to the largely increased surface and contact area. The destruction of the oligosaccharide layer may enhance the hydrolysis of cellulose through cellulase activity. The O_2-NBW reactor gained the highest ETS activity owing to the collapse of O_2-NBs and release of O_2, which can provide a micro-oxygen environment that drives the synthesis of ATP and increases the biological activity. Results from the microbial community structure analysis suggest that the micro-oxygen environment might promote the hydrolysis stage of AD. Besides, NBW addition can also enhance metabolisms of microbes including hydrogenotrophic and acetoclastic methanogens, further improving methane production from cellulose.

4. Conclusions

The present study provided a promising approach to increase methane yield from AD of cellulose through micro-oxygen environment provided by O_2-NBW. In agreement with the hypothesis, O_2-containing gas NBW supplementation improved methane production from cellulose during AD. 8%–30% increase of methane yield and 8%–14% increase of cellulose reduction were achieved from AD of cellulose with NBW addition compared to the control (DW addition). Cellulase activity, cellulose crystallinity reduction and ETS activity were also enhanced under NBW addition. O_2-containing gas NBW addition was found to enrich facultative bacteria and hydrogenotrophic methanogens. In this study, we first confirmed that O_2-NBW could provide micro-oxygen environment which has positive effects on methane production from cellulose in the lab-scale tests. Results from this work can also help us better comprehend the fundamentals involved. As seen, O_2-containing gas NBW can create micro-oxygen environment that is conductive to enhanced methane production with no chemical addition and possibly no additional energy consumption during AD. Future

research is necessary on the application potential tests at pilot- and/or full-scale to confirm its improvement on methane production from lignocellulosic biomass (e.g. corn straw, rice straw, etc.).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by JSPS KAKENHI Grant Numbers JP18H03403 and JP16H06382. Mr. Xuezhi Wang would like to thank the financial support by the China Scholarship Council (No. 201706400077) for his PhD study at University of Tsukuba, Japan.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2020.125652.

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